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# PETROLEUM TRAINING INSTITUTE, EFFURUN

SEC 114 — Past Questions & Materials

**by Renance**

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Course: SEC 114

Institution: Petroleum Training Institute, Effurun

Papers in this pack: 1

Levels: ND1

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# Paper 1 of 1 — PHYSICS SEC 114 ENVIRONMENTAL SCIENCE

Level ND1 • Second Semester • Handout

DATE: EXPERIMENT

NO: ONE TITLE: VERIFICATION OF SNELL'S LAW

AIM: To verify Snell's law using rectangular glass block.

To determine the index of refraction of a glass block

APPARATUS: Rectangular Glass block, 4 Optical pins, Drawing Board,

Tracing paper, Pencil, Protractor, Rule, Eraser

DIAGRAM:

THEORY: Snell's law states that the ratio of the sine of angle of incidence to the sine of angle of refraction is equal to a constant for a given media.

$\sin i$

$= n$ , a constant for a given pair of media

$\sin r$

PROCEDURE:

- Trace the outline of the given rectangular glass block on a tracing paper and label the outline ABCD,
- Draw a line (normal) perpendicular to line AB at E,
- Measure an angle of incidence in degrees from the normal,  $i = 20^\circ$
- Place two optical pins P and Q on the incidence ray and replace back the rectangular block on its outline,
- View the two pins, P and Q through the glass block and insert the other pins R and S so that they appear to be in line with the images of P and Q as shown in the diagram.
- Remove the rectangular block, draw a line to join R and S to meet the line CD at F,
- Draw another normal N<sub>2</sub> at F, also draw a line to join E and F
- Measure the angle of refraction made by line EF with N<sub>1</sub>
- Tabulate your observation and repeat the procedure with the values of  $i = 30^\circ, 40^\circ, 50^\circ, 60^\circ$  and  $70^\circ$

RESULTS

$S / \sin i$

$= n$

NO sinr

i r sin i sin r

1 200

2 300

3 40

4 50

5 60

6 70

QUESTIONS:

- ☐ Plot a graph of sin i against sin r
- ☐ Determine the slope of the graph, and comment on the slope
- ☐ List the possible sources of error while carrying out the experiment
- ☐ List the precautions taken to minimise the errors
- ☐ make your conclusion

DATE EXPERIMENT NO:

TWO

TITLE: TRIANGULAR PRISM

AIM: To determine the refractive index of a triangular prism

APPARATUS: Triangular prism, 4 Optical pins, Tracing paper, Drawing board, Protractor, Pencil, Eraser, Rule.

DIAGRAM:

THEORY: The prism deviate the incident ray through an angle known as angle of deviation. This is the angle between the incident ray and the emergent ray of light passing through a prism. The amount of deviation is determined by:

1. Refracting angle (A) of the prism
2. The angle of incidence
3. The refractive index of the material of the prism

PROCEDURE:

- ☐ Place the triangular glass prism on the tracing paper and trace its outline ABC
- ☐ Draw a normal on the line AB at E
- ☐ Measure the angle of incidence in degrees from the normal starting with 300
- ☐ Fix two optical pins P1 and P2 on the line to define the incidence ray and replace the prism back on its outline
- ☐ View the images of P1 and P2 from the face AC, search and

locate the most appropriate straight line for the images of P1 and P2

□ Fix the other two pins P3 and P4 to be in a straight line with the images of P1 and P2

□ Remove the prism, join the points P3 and P4 to cut AC at F and extend it with dotted line

□ Draw a normal at F to allow for the measurement of the angle of emergence (  $e$  )

□ Measure the angle of deviation (  $D$  ) ;it's the angle formed where incidence ray meet with emergence ray, angle of emergence and tabulate your readings

□ Repeat the procedure with  $i=400,500,600,700$  and  $800$

RESULTS/TABLE:

S/ Angle of Angle of Angle of  
NO incidence deviation emergence

(i) (D) (e)

1

2

3

4

5

6

QUESTIONS:

1. Plot a graph of  $D$  against  $i$
2. Calculate the refractive index of the triangular prism

1

$\sin ( A + D )$

2

$\sin A =$

g

1

$\sin A$

2

3. List the precautions taken so as to ensure accurate result.

4. List the sources of error which can alter the result obtained.

5. Draw your conclusion.

DATE: EXPERIMENT NO:

THREE

TITLE: VERIFICATION OF LAWS OF REFLECTION

AIM: To verify the laws of reflection using a plane mirror

APPARATUS: Plane mirror, optical pins, wooden block, tracing paper, Drawing board, Protractor, Pencil, Rule.

DIAGRAM:

THEORY: The laws of reflection states that;

1. The incident ray, the reflected ray and the normal at the point of incidence all lies on the same plane.
2. The angle of incidence ( $\angle i$ ) is equal to the angle of reflection ( $\angle r$ ), i.e  $\theta_i = \theta_r$

PROCEDURE:

- Fix the tracing paper firmly on the drawing board and draw a straight line AB on the paper.
- Draw a normal to cut AB at O, and measure the angle of incidence  $20^\circ$  from the normal
- Draw a line of incidence and place two optical pins on the line
- With your eye from the other side of the normal view the image of P1 and P2
- Search and locate the most appropriate straight line of the image of P1 and P2, place two other pins P3 and P4 in line with the image
- Remove the pins and join their points to the normal
- Measure and record the value of the angle of reflection.
- Repeat the procedure with  $i = 30^\circ, 40^\circ, 50^\circ, 60^\circ$  and  $70^\circ$

RESULT:

S/NO

- 1
- 2
- 3
- 4
- 5
- 6

Define reflection, and briefly explain two types of reflection with their examples

What are the necessary precautions to be taken when conducting this experiment?

List the possible sources error

Draw your conclusion

DATE: ..... EXPERIMENT

NO: FOUR

TITLE: MAGNETIC FIELD

AIM: To determine the magnetic field pattern of a bar magnet using iron filings

APPARATUS: Two no's of bar magnet, iron filings, cardboard paper, pencil, and drawing board

DIAGRAM:

(1) Magnetic field of two adjacently placed bar magnet (like poles)

N

N

(2) Unlike poles:

N

S

PROCEDURE:

□ Place the two bar magnets at some distance apart on the drawing board at some distance apart with like poles facing each other

□ Place your card board paper on top of the bar magnets

□ Gently sprinkles iron filings on the card board paper surrounding the bar magnets and spread the iron filings evenly around the bar magnets

□ Use a sharp pencil to trace the pattern of the magnetic field form by the iron filings

□ Remove the iron filings, determine and label your neutral point

For Unlike Poles use the same procedure, now with unlike poles placed adjacently

QUESTIONS

1. Define the magnetic field of a bar magnet
2. What did you understand by the term Neutral Point around a magnet
3. State the precautions to be taken during experiment
4. State the sources of error
5. Draw your conclusion

DATE: ... EXPERIMENT

NO: FIVE

TITLE: FOCAL LENGTH OF A CONVERGING LENS

AIM: To determine the focal length of a converging lens using optical bench

APPARATUS: converging lens, source of light (candle), screen, mirror stand,

Rule

DIAGRAM:

Object convex lens screen

THEORY: The focal length ( $f$ ), of a lens is related to the object distance ( $x_o$ ), and image distance ( $x_i$ ). The object distance is the distance between the object and the lens, the image distance is the distance between the lens and the image onto a screen.

$\frac{1}{f} = \frac{1}{x_o} + \frac{1}{x_i}$

$f \quad x_o \quad x_i$

PROCEDURE:

- Set the apparatus as shown in the diagram above and Light up the candle
- measure the object distance  $x_o = 30 \text{ cm}$ , i.e. the distance from the object to the lens
- Adjust the screen to and fro until the image of the flame is focused clearly but inverted on the screen
- Now measure the image distance  $x_i$ , i.e. the distance from the lens to the screen
- Repeat the procedure with different values of  $x_o = 40 \text{ cm}$ ,  $50 \text{ cm}$ ,  $60 \text{ cm}$ ,  $70 \text{ cm}$  &  $80 \text{ cm}$  in each case measure and record your observation

QUESTIONS:

Observation /Table

S/  $x_o$  ( cm )  $x_i$  ( cm )  $\frac{1}{x_o} + \frac{1}{x_i}$

( cm ) ( cm )  $\frac{1}{f} = \frac{1}{x_o} + \frac{1}{x_i}$   $f = M$

NO  $x_o \quad x_i \quad x_o$

- 1
- 2
- 3
- 4
- 5
- 6

1 1

Plot a graph of  $x$  on  $y$  –axis and  $x$  on  $x$  –axis, from the graph

io

determine the intercept of the graph

Define the following, with respect to converging lens;

i. Principal focus (F), ii. Focal length (f) iii Magnification

iv. Principal axis, v. Optical Centre (C)

An object 5cm high is placed at a distance 12cm from a convex lens of focal length 8cm. Calculate the position, size and nature of the image.

List the possible sources of error during the experiment.

List the necessary precautions taken to minimise the errors.

State your conclusion.

DATE...

EXPERIMENT NO: SIX

TITLE: DETERMINATION OF DENSITY

AIM: To determine the density of various objects

APPARATUS: Wooden block, cylinder, small black pipe, a sphere, beam balance, vernier calliper, meter rule.

DIAGRAM: Sketch a pictorial diagram of each object and indicate its dimensions

THEORY: Density ( $\rho$ ) is defined as the mass per unit volume of the material

mass

i . eDensity =

volume

The unit of density is  $\text{kg m}^{-3}$ , smaller units are  $\text{g cm}^{-3}$

PROCEDURE:

□ Use a beam balance to measure the mass of each object and record it in the table below

□ Measure and record the dimension of each object and record it in the table below, block(length, width and height,  $l \times b \times h$ ), cylinder/black pipe(diameter and length  $V = \pi r^2 l$ ),

4

sphere(diameter  $V = \frac{4}{3} \pi r^3$ )

3

□ Calculate the volume of each object and record it in the observation table



□ Use the values obtain for mass and volume to calculate the density of each of the components, also record it in the observation table below

OBSERVATION/TABLE:

Obj Mass( Volum Density(gcm- Density(gcm-3) %error  
ct g) e 3

Standard

Experimenta  
(cm<sup>3</sup>)

I

experimental value –standard value

Percentage error =  $\frac{\text{experimental value} - \text{standard value}}{\text{standard value}} \times 100 \%$

standard value

QUESTIONS:

□ Define the term density and differentiate it from the relative density

□ Calculate the density of a substance having a mass of 30g and a volume of 8cm<sup>3</sup>; express your answer in terms of meters and kilograms.

□ List the sources of error likely to interfere with the result obtained

□ State the precautions taken to minimise those errors

□ Conclusion

DATE..... EXPERIMENT

NO: SEVEN

TITLE: SPECIFIC HEAT CAPACITY

AIM: To determine the specific heat capacity of a solid (brass and aluminium)

APPARATUS: Copper calorimeter, red cork, thermometer, beam balance, water heater, water, standard masses, hot water,

DIAGRAM:

DEFINITION:

A calorimeter is a device that can be used to determine the specific heat of a metal; it is required so as to reduce the effects of the environment which ideally can be neglected

PROCEDURE:

□ Use a beam balance to measure and record the mass of the following;

Solid (brass or aluminium), empty calorimeter cup,

calorimeter with water, water and record them

□ Suspend the solid in boiling water and allow it to boil for about 10 minutes so that it attains the temperature of the boiling water ( $T_{\text{boiling}} = T_{\text{solid}}$ )

□ Measure and record the temperature of the water inside the calorimeter cup, this serves as  $T_{\text{water}}$

1 &  $T_{\text{cup}}$

1. The volume of

water used should allow for total submersion of the hot solid.

□ Remove the hot solid from the boiling water, and transfer it to the calorimeter and cover it immediately.

□ Lightly stir the ingredients, with thermometer inserted, observe and record the highest temperature attained.

$T_{\text{solid}}$

2 =  $T_{\text{water}} = T_{\text{cup equivalent}}$

2 =  $T_2$

□ Calculate the specific heat of the solid material used from the following statement of the conservation of energy:

$Q_{\text{solid}} + Q_{\text{water cup environment}}$

lost +  $Q_{\text{gained}} + Q_{\text{gained}} + Q_{\text{lost}} = 0$

$\Rightarrow m_{\text{solid}} c_{\text{solid}} (T_2 - T_1) + m_{\text{water}} c_{\text{water}} (T_2 - T_1) + m_{\text{cup}} c_{\text{cup}} (T_2 - T_1) = 0$

solid solid water water cup cup

OBSERVATION/TABLE:

EXPERIMENTAL DATA

Identity of solid

Mass of solid

$m_{\text{solid}}$

Mass of water

$m_{\text{water}}$

Mass of cup

$m_{\text{cup}}$

Initial temperature of cork

solid

$T_1$

Initial temperature of water

$T_{\text{water}}$

1

Initial temperature of cup

cup

T<sub>1</sub>

Equilibrium Temperature  $T_{eq} =$

$T_{solid}$

$2 = T_{2water} = T_{cup}$

2

Specific heat of water,  $c_{water} 4.184 \text{ J/g}^\circ\text{C}$

Specific heat of cup,  $c_{cup} 0.385 \text{ J/g}^\circ\text{C}$

Specific heat of solid  $c_{solid}$ ,

using the above equation

Specific heat from the table of

constants/standard

Percentage error

QUESTIONS:

Define the specific heat capacity and differentiate it from heat capacity

State all the possible sources of error

What precautions do you take to minimise them

Draw your conclusion

DATE..... EXPERIMENT

NO: EIGHT

TITLE: HEAT OF FUSION OF ICE

AIM: To determine the latent heat of fusion of ice

APPARATUS: Calorimeter, ice block, water (warmish),  
thermometer, beam balance, standard masses, water heater

DIAGRAM:

THEORY: The heat of melting, also known as the latent heat of fusion is the amount of heat required to change the ice to water without changing the temperature.

Heat of fusion ( $Q$ ) =  $mL_f$

Where  $m$  is the mass of the solid, &  $L_f$  is its latent heat

PROCEDURE:

□ Record the specific heat and mass of the inner calorimeter cup.

□ Record the air temperature, and warm some water to about  $10^\circ\text{C}$  above the room temperature (warmish) and pour 60 ml into the calorimeter, and place the lid on it.

- Determine the mass of the calorimeter and water. Record the temperature of the water after stirring.
- Place several small pieces of ice in an empty cup and determine the mass of the ice and cup.
- Remove the lid of the calorimeter and place the ice one piece at a time into the warm water. Replace the cover and mildly shake or stir its ingredients. Continue adding ice until the temperature falls well below room temperature.
- Measure the mass of the calorimeter cup, water and now melted ice.
- Calculate the heat of fusion by solving for  $L_f$  in the equation below:

$$m_{ice} L_f + m_{ice} c_{ice} (T_{ice} - 0) + m_w c_w (T_2 - T_1) + m_{cup} c_{cup} (T_2 - T_1) = 0$$

OBSERVATION/TABLE:

EXPERIMENTAL DATA

S/NO

1 Room Temperature

2 Temperature of warm water:

$w_{cup}$

$T = T$

1 1

3 Equilibrium temperature of

$w_{cup}$

water:  $T_2 = T_2$

4 Mass of empty cup:  $m_{cup}$

5 Mass of warmish water and cup:

$m_w + m_{cup}$

6 Mass of warmish water:  $m_w$

7 Mass of melted ice and water:

$m_{ice} + m_w$

8 Mass of ice:  $m_{ice}$

9 Specific heat of cup:  $c_{cup}$

10 Heat of fusion of ice from data:

$L_f$

11 Accepted value of heat of fusion

12 Percentage error

List the sources of error

State the precautions taken to minimize the errors

Conclusion

DATE

EXPERIMENT NO: NINE

TITLE: NEWTON'S LAW OF COOLING

AIM: To verify Newton's law of cooling

To draw the cooling curve

APPARATUS: Copper calorimeter with a stirrer, thermometer, water heater, water, stopwatch.

DIAGRAM

DEFINITION/THEORY

Newton's law of cooling states that the rate of cooling of a body is directly proportional to the temperature difference between the body and the surroundings.

The law holds good only for a small difference of temperature.

Loss of heat by radiation depends on the nature of the surface and the area of the exposed surface.

Newton's Law of Cooling states that the rate at which convection cools a hot object is proportional to the temperature difference between the hot object and the ambient (room) temperature.

$dT/dt$  is proportional to  $(T - T_{\text{ambient}})$ .

This is stated mathematically as  $dT/dt = -k(T - T_{\text{ambient}})$

Since this cooling rate depends on the instantaneous temperature (and is therefore not a constant value), this relationship is an example of a 1st order differential equation.

The constant  $k$  in this equation is called the cooling constant.

The solution to this differential equation is

PROCEDURE

The calorimeter is filled 2/3rd with the given liquid and is heated to a temperature of  $80^{\circ}\text{C}$ . This liquid will act as a hot body which is subjected to cooling. The thermometer is inserted into the calorimeter. When the temperature reading is  $70^{\circ}\text{C}$  the stopwatch is started. The time readings are noted for every  $5^{\circ}\text{C}$  fall of temperature up to the room temperature. The readings are tabulated. A graph is drawn with temperature  $\theta$  along Y axis and time ( $t$ ) along X axis,  $d\theta/dt$  is found by taking slopes to tangents drawn at various temperatures on the cooling curve.

Hence Newton's law of cooling is verified.

#### TABLE OF READINGS

S/ TEMPERATURE TIME  $t$ , Ambient temp(  $^{\circ}\text{C}$  )

N  $\theta$ , (  $^{\circ}\text{C}$  ) (min)

Plot a graph of temperature ( $^{\circ}\text{C}$  ) against time (min) and determine the slope at various tangents of the graph

List sources of error while conducting the experiment

List precautions to minimise the errors

Conclusion

DATE

EXPERIMENT NO: TEN

TITLE: OPERATION OF A HEAT ENGINE

AIM: To understand and demonstrate the operation of a heat engine using Pasco Scientific Heat engine/Ideal Gas Law

Apparatus

APPARATUS: Pasco Scientific Heat Engine/Ideal Gas Law

Apparatus set of standard mass, container of hot water, container of cold water

DIAGRAM

Definition/Theory: a heat engine is a device that converts thermal energy to other useful forms, such as electrical and mechanical energy.

Pasco Scientific Heat engine/Ideal Gas Law Apparatus consist of air inside cylinder that expands when an attached can is immersed in hot water